10.9 Symmetry Factors

In quantum field theory, symmetry factors correct for overcounting in perturbative expansions. This note supplements the discussion in lectures: first the definition and three standard examples, then worked examples in φ^3 theory.

Origin and Definition

Symmetry factors arise because several *Wick contractions*, or permutations of vertices and lines, can produce the same term. The redundancy is measured by the **automorphism group** of the diagram: all relabelings of vertices and of the line ends attached to them (permutations, rotations, reflections) that leave the diagram unchanged. The **symmetry factor** is the order of this group, $S = |\mathrm{Aut}(G)|$, and each diagram is weighted by $1/S$.

Three Examples

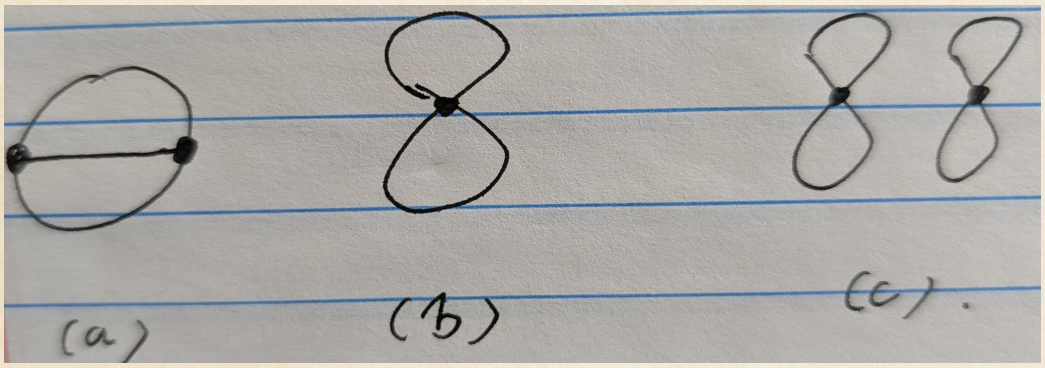


Diagram (a) is in φ^3 theory; (b) and (c) are vacuum diagrams of φ^4 theory, where each vertex has four lines.

- **The "sunset" diagram in φ^3 theory** (a): two vertices connected by three lines.
  - Symmetries: vertex swap ($\mathbb{Z}_2$) and permutations of the three lines (S_3).
  - $|\mathrm{Aut}(G)| = 2 \times 6 = 12 \Rightarrow S = 12$.
- **The vacuum "figure-eight" in φ^4 theory** (b): one vertex with two loops.
  - Symmetries: swapping the loops ($\mathbb{Z}_2$) and flipping each loop ($\mathbb{Z}_2 \times \mathbb{Z}_2$).
  - $|\mathrm{Aut}(G)| = 2 \times 2 \times 2 = 8 \Rightarrow S = 8$.
- **Two identical disconnected pieces** (c): two copies of the figure-eight.

- Symmetries: swapping the two pieces ($\mathbb{Z}_2$) and the symmetries of each piece.
- $|\mathrm{Aut}(G)| = 2 \times (8 \times 8) = 128 \Rightarrow S = 128$.

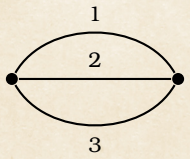
Worked Examples in φ^3 Theory

Recall how we "got rid of" the factor

$$\frac{1}{V!}\left(\frac{1}{3!}\right)^V \frac{1}{P!}\left(\frac{1}{2!}\right)^P$$

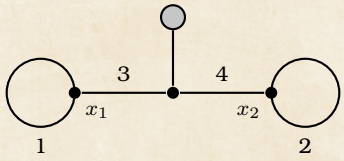
in the expansion of $Z(J)$ (Srednicki §9). We treated the V vertices and the P propagators as identical and exchangeable, which gives $V!\,P!$ rearrangements. The three derivatives $\left(\frac{1}{i}\frac{\delta}{\delta J}\right)^3$ of each vertex, acting on the factors $\frac{1}{2}J\Delta J$, then supply another $(3!)^V(2!)^P$ possibilities, cancelling the factor exactly. But this overcounts whenever some of those rearrangements give back the same term. The overcounting is the symmetry factor S, and each diagram is weighted by $1/S$. Shaded circles below are sources.

Example A: the sunset, by counting



(1) The $(3!)^2$ permutations of the two $\left[\frac{1}{i}\frac{\delta}{\delta J}\right]^3$ give only $3!$ different products Δ^3 if one distinguishes the propagators. That is an overcounting by $3!$.
(2) The $(2!)^3$ flips of the propagators are already equivalent to exchanging the two vertices, which was counted in $V! = 2!$. That is an overcounting by 2.
(3) The total overcounting is $2 \times 3! = 12$, so $S = 12$.
(4) The same number follows from the symmetries of the diagram, as for diagram (a) above: the three propagators can be permuted ($3!$), and the two vertices can be exchanged by a reflection (2).

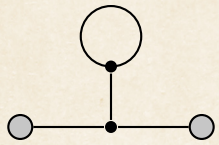
Example B: two tadpoles on a line, one source



(1) The loops 1 and 2 are $\Delta(x_1 - x_1)$ and $\Delta(x_2 - x_2)$. Flipping the ends of a loop gives back the same term; there is no $\Delta(x - y) \to \Delta(y - x)$ to tell the two orders apart. So the weight is overcounted by 2×2.

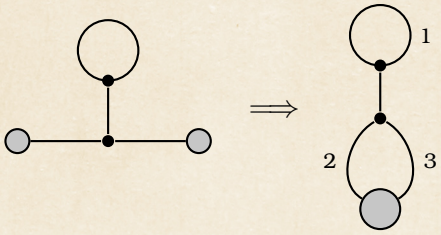
(2) Exchanging propagators 3 and 4 together with the vertices x_1 and x_2 gives the same process. It only changes how the functional derivatives are arranged and produces no new term, so it overcounts by another 2.
(3) The symmetry factor is $S = 2 \times 2 \times 2 = 8$.
(4) By symmetry: a left–right reflection, and an up–down reflection of each of the two loops, give $S = 2 \times 2 \times 2 = 8$ again.

Example C: a tadpole on a line between two sources



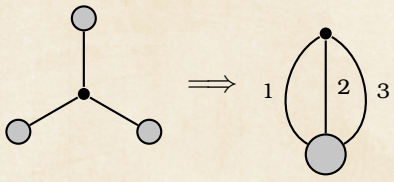
(1) The loop gives an overcounting of 2.
(2) The left–right reflection gives another 2.
(3) $S = 2 \times 2 = 4$.

For several sources there is a useful *deformation method*. Since the sources are identical, deform the diagram so that all of them meet in one shaded blob. The blob is not a vertex: any number of lines may end on it. Then read off the symmetries of the deformed diagram. For Example C:



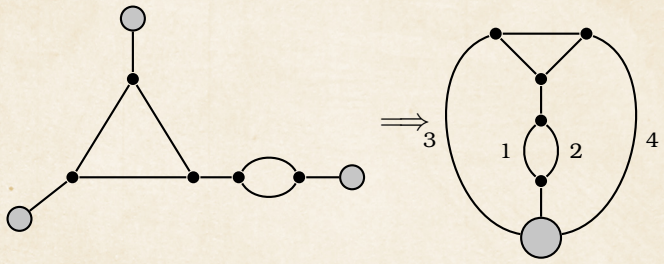
Propagator 1 gives a factor of 2, and propagators 2 and 3 can be exchanged ($2!$). So $S = 2 \times 2 = 4$.

Example D: three sources on one vertex



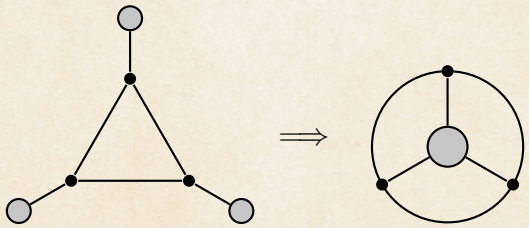
The three propagators can be permuted in every way, so $S = 3! = 6$.

Example E: a triangle with a bubble on one leg



After the deformation one can exchange $1 \leftrightarrow 2$ (the two lines of the bubble) and $3 \leftrightarrow 4$ (a reflection of the triangle that fixes the dressed corner). So $S = 2 \times 2 = 4$.

Example F: a triangle with three bare legs



The deformed diagram has a cyclic symmetry of order 3 (rotations by $120°$ and $240°$) and a reflection. The three reflections, one through each corner, are not independent: each is a rotation times one reflection. So $S = 3 \times 2 = 6$.

Why $1/S$: Terms, Diagrams and Relabelings

At the sunset (the theta of LecNote8 p. 4) the margin asks: "we only have one possibility! How can I count it $2! \times 3!$ times?" And where did the $1/3!$ of each vertex and the $1/2$ of each line go? With $Z_g = 1$ and $\mathcal{L}_{\rm ct}$ dropped, LecNote8 p. 3 expands

$$Z_1(J) \propto \sum_{V=0}^{\infty} \frac{1}{V!}\Big[\frac{ig}{3!}\int d^4x \Big(\frac{1}{i}\frac{\delta}{\delta J(x)}\Big)^3\Big]^V \times \sum_{P=0}^{\infty} \frac{1}{P!}\Big[\frac{i}{2!}\int d^4y\, d^4z\, J(y)\Delta(y-z)J(z)\Big]^P, \quad (10.9.1)$$

with Srednicki's Δ, $(-\partial^2 + m^2)\Delta = \delta^4$. Each of the $3V$ derivatives hits a different one of the $2P$ factors J; the $E = 2P - 3V$ left over are sources. A **term** is one such assignment, and there are $(2P)!/E!$ of them. A **diagram** D is the class of all terms that draw the same picture.

Renaming the V vertex variables, permuting the three derivatives at each vertex, permuting the P propagator factors and swapping the two ends of

each (Δ is even) leave Eq. (10.9.1) unchanged. These relabelings form a group R of order

$$|R| = V!\,(3!)^V\,P!\,(2!)^P, \tag{10.9.2}$$

exactly the inverse of the prefactor: that is where the $1/3!$ and the $1/2$ go. A relabeled term has the same value (it is the same integral with renamed variables), and two terms draw the same picture exactly when a relabeling carries one into the other. So the terms of D form one orbit of R, and by the orbit–stabilizer theorem the orbit has $|R|/|\mathrm{Stab}_D|$ terms, where Stab_D, the relabelings that map a term to itself, is $\mathrm{Aut}(D)$ of "Origin and Definition". Times the prefactor $1/|R|$, the diagram contributes its value divided by $|\mathrm{Aut}(D)| = S$. So $1/S$ is not put in by hand: it is what is left of the Taylor coefficients once identical terms are added up.

The count for $V = 2$**,** $P = 3$**.** The six derivatives hit the six J's in $6! = 720$ ways. For the sunset, vertex 1 sends its legs to three different lines ($3!$) and picks an end of each (2^3), and vertex 2 fills the other ends ($3!$): $3! \cdot 2^3 \cdot 3! = 288$ terms. For the dumbbell, one leg at each vertex goes to the bar ($3 \cdot 3$), the lines take the roles bar, loop at vertex 1, loop at vertex 2 ($3!$), and the ends are filled (2^3): 432 terms, and $288 + 432 = 720$. With $|R| = 2! \cdot 6^2 \cdot 3! \cdot 2^3 = 3456$, the coefficients are $288/3456 = 1/12$ and $432/3456 = 1/8$. The margin's $2! \times 3!$: of the 3456 relabelings, 12 map a given sunset term to itself (a permutation of the lines, with or without a swap of the vertices, each with the matching legs and ends); the other 3444 carry it to the other 287 sunset terms, 12 to each.

A Completeness Test: the Sum Rule

"How do I know that my list of diagrams (LecNote8 pp. 4–5) is complete, and every S right?" Every term belongs to exactly one diagram, so $\sum_D |R|/S_D = (2P)!/E!$. With $2P = 3V + E$ and $(2P)!/(P!\,2^P) = (2P-1)!!$,

$$\sum_D \frac{1}{S_D} = \frac{(3V+E-1)!!}{V!\,6^V\,E!}, \tag{10.9.3}$$

summed over all diagrams with V vertices and E sources, connected or not. The numerator is the number of Wick pairings of the $3V$ legs and E sources. A forgotten diagram or a wrong S breaks the equality. The table below checks the cases of LecNote8: in every row the last column equals the right side, e.g. $15/(2 \cdot 36) = 5/24$ for $(0,2)$ and $10395/(24 \cdot 6^4) = 385/1152$ for $(0,4)$.

The disconnected S are products over the pieces, with an extra $2!$ for two identical pieces (next subsection). The other unnamed connected ones: the bubble on a line at $(2,2)$; at $(1,3)$, a source on a triangle with one side doubled, and a source line through a bubble into a tadpole; at $(0,4)$, the tetrahedron (24), three tadpoles on one vertex (48), a

(E,V)	connected S	disconnected S	$\sum_D 1/S_D$
$(0,2)$	12 (sunset), 8 (dumbbell)	—	$5/24$
$(1,1)$	2 (tadpole)	—	$1/2$
$(3,1)$	6 (Example D)	4	$5/12$
$(2,2)$	4 (Example C), 4	8, 16, 24	$35/48$
$(1,3)$	8 (Example B), 4, 4	16, 24	$35/48$
$(0,4)$	24, 48, 16, 16, 8	128, 96, 288	$385/1152$

THE SUM RULE FOR THE DIAGRAMS OF LECNOTE8.

tadpole–bubble–tadpole chain and a ring of two bubbles (16 each), and a tadpole on a triangle with one side doubled (8).

DISCONNECTED DIAGRAMS AND $Z = e^{iW}$

LecNote8 pp. 7–8: why do n identical pieces carry $1/n!$, and why should the *logarithm* of Z be connected? Let D consist of n_I copies of each connected diagram C_I. An automorphism maps every connected piece onto an identical one, so it permutes identical copies and acts on each:

$$|\mathrm{Aut}(D)| = \prod_I n_I!\, S_I^{n_I}. \tag{10.9.4}$$

Diagram (c) above is one case ($2 \cdot 8 \cdot 8$); in φ^3 theory, two sunsets give $2! \cdot 12^2 = 288$, two dumbbells $2! \cdot 8^2 = 128$, a sunset and a dumbbell $12 \cdot 8 = 96$. With C_I also standing for the value of the connected diagram, $1/S_I$ included, D contributes $\prod_I C_I^{n_I}/n_I!$: the extra factor S_D of LecNote8 p. 7 is $\prod_I n_I!$, not its inverse, and it divides. Summing over every n_I independently,

$$Z_1(J) \propto \sum_{\{n_I\}} \prod_I \frac{C_I^{n_I}}{n_I!} = \exp\Big(\sum_I C_I\Big), \tag{10.9.5}$$

and once the vacuum diagrams are divided out (p. 8; what they are and why this is allowed is in §10.10), $Z = e^{iW}$ with iW the connected diagrams with sources (p. 9).

This is no accident of the algebra. In probability the logarithm of a generating function generates the cumulants, and a Gaussian of zero mean has only one, its variance; correspondingly $\ln Z_0 = \frac{i}{2}\int J\Delta J$ is a single connected diagram. The physical reason is **cluster factorization**: for sources J_1, J_2 far apart in space, the two experiments cannot influence each other, so $Z(J_1 + J_2) \to Z(J_1)Z(J_2)$ and W must be additive. Products of separated pieces cannot appear in W; its terms that involve both J_1 and J_2 must die off as the sources are pulled apart. A connected diagram does, because it contains propagators across the gap, and these fall off like e^{-mr} at spacelike distance r. In the free theory the only cross term is $i\int J_1 \Delta J_2$; for point sources j_1, j_2 held on for a long time T it is $ij_1j_2T\, e^{-mr}/4\pi r$, since $\int dt\, \Delta(t,\mathbf{r}) = e^{-mr}/4\pi r$, the Yukawa potential.

Labelled External Lines

LecNote9 p. 3 finds the $4!$ ways to apply $\delta_1\delta_2\delta_3\delta_4$ to fall into "3 groups ... each with 8 identical diagrams" and calls the cancellation of S "a general result for TREE-level diagrams"; rule 8 on p. 7 keeps a factor for loops. Why, and when does one survive?

$\langle 0|T\varphi(x_1)\cdots\varphi(x_E)|0\rangle$ comes from $\frac{\delta}{i\delta J(x_1)}\cdots\frac{\delta}{i\delta J(x_E)}$ acting on Z. On a diagram with E sources and weight $1/S$, the derivatives remove the sources in $E!$ ways, each putting the labels $x_1,\ldots,x_E$ on the ends. Two assignments give the same labelled diagram exactly when an automorphism relates them, and those that fix an assignment fix every source; call their number $S_{\rm lab}$. By orbit–stabilizer, each labelled diagram arises $S/S_{\rm lab}$ times, there are $E!\,S_{\rm lab}/S$ of them, and each has weight

$$\frac{1}{S}\cdot\frac{S}{S_{\rm lab}}=\frac{1}{S_{\rm lab}}. \tag{10.9.6}$$

These orbits are the "topologically inequivalent" diagrams of LecNote9's rule 2 (p. 6).

Trees have $S_{\rm lab}=1$. Removing a vertex of a tree leaves three branches, each ending at an external point, so every vertex lies on the path between two external points. Paths in a tree are unique, so an automorphism that fixes the external points fixes every vertex and line, and (with no loops or multiple lines) every leg. The H of LecNote9 p. 3 has $S=8$, the swaps in the page's parenthesis, so $4!=3\times 8$: three labelled diagrams, the s, t and u channels, each with weight 1.

Loops can keep a symmetry with the ends fixed. The bubble on a line goes from $S=4$ (swap the bubble's lines; reflect end to end) to $S_{\rm lab}=2$, the $\frac{1}{2}$ of the one-loop self-energy (LecNote12 p. 2); Example C also goes from 4 to 2. In $2\to 2$ at one loop the box goes from 8 to 1 (three labelled boxes), the triangle at one vertex from 4 to 1 (six), and the bubble on the internal line from 16 to 2 (three); §10.11 lists all the one-loop $2\to 2$ diagrams. That is rule 8: a factor survives only from a loop symmetry that leaves every external point in place.

Zero Dimensions: the Vacuum Sums as a Series

Does the sum over diagrams converge? The simplest place to ask is zero dimensions, where the field is a single number, $\Delta=1$ and there are no integrals, so a diagram is worth $g^V J^E/S_D$. Expanding

$$Z(g,J)=\frac{1}{\sqrt{2\pi}}\int d\varphi\, e^{-\varphi^2/2+g\varphi^3/6+J\varphi} \tag{10.9.7}$$

with $\langle\varphi^n\rangle=(n-1)!!$ gives the right side of Eq. (10.9.3) as the coefficient of $g^V J^E$, so $Z=\sum_D g^V J^E/S_D$, a formal series (the integral diverges for every

real $g \neq 0$). The vacuum sums of the table are the coefficients $c_k = (6k-1)!!/\big[36^k(2k)!\big]$ of g^{2k} in

$$Z(g,0) = 1 + \tfrac{5}{24}g^2 + \tfrac{385}{1152}g^4 + \cdots, \qquad \ln Z(g,0) = \tfrac{5}{24}g^2 + \tfrac{5}{16}g^4 + \cdots,$$

and the logarithm keeps only the connected ones: the disconnected $\frac{1}{288} + \frac{1}{128} + \frac{1}{96} = \frac{1}{2}(\frac{5}{24})^2$ is the second-order term of the exponential. The pairings outrun the prefactor, $c_{k+1}/c_k \to 3k/2$: factorial growth and zero radius of convergence, as for φ^4 in §10.17, but with all $c_k > 0$ where φ^4's alternate.

Summary

The contribution of a diagram to the amplitude is

$$\text{Contribution} = \frac{1}{S} \times (\text{couplings}) \times (\text{integrals}).$$

Symmetry factors ensure that each distinct term is counted exactly once. Errors in S propagate to observables such as cross sections, so getting them right matters. I directly write down the symmetry factors of many diagrams in my lecture notes and provide additional notes explaining different ways to get them. There is ultimately one method: carry out the functional-derivative expansion of the path integral. There, you won't need to introduce symmetry factors, but rather count how many ways there are to get the identical expression.